

Adaptive Noise-Robust Offline Voice-Controlled Microcontroller using Tiny-ML

Introduction

Voice interfaces are no longer futuristic — they're part of our daily lives. From asking smart speakers to play music, to controlling appliances with a simple command, voice has become a natural way to interact with machines. But here's the catch: most of these systems rely on **cloud servers** or **powerful processors**. That means they're not suitable for small, battery-powered devices. They introduce **delays**, raise **privacy concerns**, and consume **too much energy**.

This project tackles those challenges head-on. By leveraging **TinyML**, we bring intelligence directly to microcontrollers. Instead of sending audio to the cloud, the device itself listens, understands, and acts. We built an **adaptive noise-robust offline voice controller** using ESP32. It listens for the keyword "RED," filters out background noise, and lights up an LED — all in real time, without internet.

Abstract

The system integrates **signal processing** and **machine learning** into a compact, low-power design. An **INMP441 microphone** captures audio, which is refined using **adaptive noise estimation** and **voice activity detection (VAD)**. Features such as **Mel-Frequency Cepstral Coefficients (MFCCs)** are extracted and passed to a lightweight keyword spotting model trained with **Edge Impulse**.

When the keyword is detected, the ESP32 triggers an LED and displays classification scores on an OLED screen. The design emphasizes **privacy, efficiency, and robustness**, making it ideal for IoT devices, smart homes, healthcare tools, and industrial automation.

Problem Statement

Voice-controlled systems are powerful but face four big hurdles in microcontroller environments:

Adaptive Noise-Robust Offline Voice-Controlled Microcontroller using Tiny-ML

- **Latency:** Cloud-based systems introduce delays in command execution.
- **Privacy Risks:** Audio data sent to external servers may expose sensitive information.
- **Energy Inefficiency:** Continuous connectivity drains battery life.
- **Noise Sensitivity:** Real-world environments often contain background noise, reducing recognition accuracy.

The challenge is to design a **noise-robust, offline voice-controlled system** that can operate reliably on microcontrollers, ensuring **low latency, privacy, and energy efficiency** while adapting to noisy conditions.

Components

- **ESP32 / Nano 33 BLE** → the microcontroller brain.
- **INMP441 I2S Microphone** → captures audio signals with high fidelity.
- **OLED Display (SSD1306)** → provides user feedback on classification results.
- **LED / Relay / Motor** → controlled outputs for actuation.
- **5V USB / Battery Supply** → portable power source.
- **Breadboard + Jumper Wires** → for prototyping connections.

Project Goals

- Build a **voice command recognition system** that works offline.
- Deploy a **TinyML keyword spotting model** on microcontrollers.
- Handle **noise adaptively**, ensuring reliability in real environments.
- Prioritize **privacy, low latency, and energy efficiency**.
- Keep the system **compact, affordable, and scalable**.
- Demonstrate potential for **multi-command expansion** and diverse applications.

Adaptive Noise-Robust Offline Voice-Controlled Microcontroller using Tiny-ML

Block diagram

Microphone (INMP441) → Adaptive Noise & VAD → TinyML Voice Recognition → MCU Control → Output (LED/Relay)

Methodology

1. **Audio Capture:** The microphone records raw audio frames.
2. **Noise Filtering:** Adaptive thresholds + VAD remove silence and background hum.
3. **Feature Extraction:** MFCCs represent speech patterns.
4. **Inference:** Edge Impulse classifier checks for the keyword “RED.”
5. **Decision Logic:** If confidence > 70%, LED turns ON.
6. **Feedback:** OLED shows peak values, scores, and LED status.
7. **Efficiency:** Silent frames are skipped to save power.
8. **Firmware Integration:** Code integrates I2S drivers, OLED libraries, and Edge Impulse SDK seamlessly.

Results & Observations

- **Accuracy:** Keyword “RED” detected with >85% confidence in normal conditions.
- **Noise Robustness:** Adaptive thresholds improved detection in noisy rooms.
- **Latency:** Commands executed in under 200 ms.
- **Power Use:** Efficient enough for battery-based IoT devices.
- **User Feedback:** OLED display made the system transparent and easy to monitor.
- **Scalability:** Framework can be extended to multiple commands.

Adaptive Noise-Robust Offline Voice-Controlled Microcontroller using Tiny-ML

Applications

- **Smart Homes** → control lights, fans, appliances offline.
- **Healthcare** → assistive voice tools for elderly or differently abled.
- **Industry** → hands-free machine operation in noisy factories.
- **Wearables** → offline voice commands for watches or trackers.
- **Agriculture** → voice-controlled irrigation pumps in rural areas.
- **Automotive** → offline in-vehicle voice control.
- **Education** → affordable voice-interactive kits for students.

Novelty

- **Adaptive Noise Handling** → adjusts sensitivity dynamically.
- **TinyML Deployment** → runs models directly on microcontrollers.
- **Privacy First** → no cloud dependency, all processing local.
- **Low Power** → energy-aware design suitable for IoT.
- **Scalable** → framework can expand to multiple commands.

Future Scope

- Add multiple keywords ("BLUE," "GREEN," etc.).
- Connect via **WiFi/MQTT** for IoT control.
- Optimize further for ultra-low-power chips.
- Support multiple languages.
- Fuse with sensors (gesture, touch) for multimodal control.
- Deploy in classrooms, hospitals, and smart cities.

Adaptive Noise-Robust Offline Voice-Controlled Microcontroller using Tiny-ML

Conclusion

This project proves that voice interfaces don't need the cloud. **By combining adaptive noise handling with TinyML keyword recognition, the ESP32 delivers**

fast, private, and reliable voice control. The system bridges the gap between modern voice technology and low-power embedded platforms, opening doors for scalable IoT solutions in homes, healthcare, industry, and education.

References

1. Banbury, C. R., Reddi, V. J., Lam, M., Fu, W., Holleman, J., Huang, X., ... & Mattina, M. (2021). *Microspeech: Embedded machine learning for keyword spotting on microcontrollers*. Proceedings of Machine Learning and Systems, 3, 219–229.
<https://proceedings.mlsys.org/paper/2021/file/2a79ea27c2798c2f7f5f0f9d6d3a9f2d-Paper.pdf> ([proceedings.mlsys.org](https://proceedings.mlsys.org/paper/2021/file/2a79ea27c2798c2f7f5f0f9d6d3a9f2d-Paper.pdf) in Bing)
2. Edge Impulse Inc. (2022). *Edge Impulse SDK documentation*. Retrieved from <https://docs.edgeimpulse.com>
3. Lane, N. D., Bhattacharya, S., Mathur, A., Georgiev, P., Forlivesi, C., Kawsar, F., & Seneviratne, A. (2015). *Squeezing deep learning into mobile and embedded devices*. IEEE Pervasive Computing, 16(3), 82–88.
<https://doi.org/10.1109/MPRV.2015.38> ([doi.org](https://doi.org/10.1109/MPRV.2015.38) in Bing)
4. Warden, P., & Situnayake, D. (2019). *TinyML: Machine learning with TensorFlow Lite on Arduino and ultra-low-power microcontrollers*. O'Reilly Media.
5. Zhang, Y., Suda, N., & Lai, L. (2017). *Hello Edge: Keyword spotting on microcontrollers*. arXiv preprint arXiv:1711.07128.
<https://arxiv.org/abs/1711.07128>

Adaptive Noise-Robust Offline Voice-Controlled Microcontroller using Tiny-ML

6. Google Patents. (2022). *Noise robust representations for keyword spotting systems (US12027156B2)*. Retrieved from <https://patents.google.com/patent/US12027156B2> ([patents.google.com in Bing](#))
7. Espressif Systems. (2023). *ESP32 technical reference manual*. Retrieved from <https://www.espressif.com/en/products/socs/esp32/resources> ([espressif.com in Bing](#))

The End